

PICK THE RIGHT KPIS

KPI	Good	Warning	Bad	Source
Ping Latency (WAN)	<70 ms	70-150 ms	>150 ms	ITU-T G.114
Packet Loss	<0.1 %	0.1-1 %	>1 %	Obkio
Jitter (VoIP)	<30 ms	30-50 ms	>50 ms	Cisco design
CPU (Router)	<60 %	60-80 %	>80 %	Vendor docs
Interface Util (95th)	<50 %	50-70 %	>70 %	Auvik

Tip: Track the 95th percentile to ignore bursty peaks.

Baseline metrics & smart alert thresholds (first half)

Why baseline?

Kentik research shows latency spikes start hurting VoIP quality once one-way delay exceeds 150ms or packet loss tops 1%. An alert set too tight floods Slack; too loose, and payroll SaaS dies silently.

Build a 14-day baseline in Zabbix

1. **Templates:** Link Template Net Network Generic SNMP to every switch and AP.
2. **Macros:** Set `{IF.ERRORS.MAX}` to 30 (errors/hour) per interface.
3. **Housekeeping:** In Administration → General → Housekeeping, keep trends for 365 days to show seasonality.

Create “Smart” triggers

```
{Template Net Generic SNMP:ifInErrors[ifDescr].change(5m)}>{$IF.ERRORS.MAX}
```

Alarms only if errors exceed baseline by the macro value – avoids flapping when a user unplugs a cable.

Percentile-based capacity alarms

Fixed “70% link utilisation” alarms flap the moment your CEO uploads a 4K video. A better tactic is to trigger on the 95th-percentile of bandwidth – a method Indian ISPs themselves use for billing. Collect five-minute SNMP counters for a fortnight, sort, drop the top 5%, and alert only if that 95th value exceeds 70 % of line rate.

Zabbix How-To

```
# Create an item prototype on the uplink port
key: net.if.in[ifHCInOctets,"{$UPLINK_PORT}"]
```

